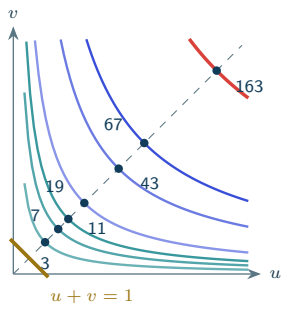
HEEGNER NUMBERS ON ONE PAGE

Euler's n^2+n+41 and the Heegner numbers

Each Heegner number is a line $u+v=1$ that misses the hyperbola $uv=k$, and it misses it modulo every prime below k.

1 / THE RULE AND THE PICTURE

$f(n)=n^2+n+k$ with $k=\frac{d+1}{4}$ is the value $k-uv$ at the lattice points $(n+1,-n)$ of the line $u+v=1$



Writing $g(x)=x^2-x+k$ we have $f(n)=g(n+1)=k-uv$ for $(u,v)=(n+1,-n)$, so the roots of f are where the line $u+v=1$ meets the hyperbola $uv=k$, and $\Delta=1-4k=-d$:

$$\boxed{d=4k-1,\qquad \text{line } u+v=1 \text{ misses } uv=k}$$

In the first quadrant the line reaches only $uv\le\frac14$, at $(\frac12,\frac12)$, so every hyperbola $uv=k$ with $k=1,2,3,5,11,17,41$ passes above it, with its vertex $(\sqrt{k},\sqrt{k})$ on the diagonal labelled by the Heegner number d.

n	0	1	2	3	4	5
(u,v)	$(1,0)$	$(2,-1)$	$(3,-2)$	$(4,-3)$	$(5,-4)$	$(6,-5)$
$41-uv$	41	43	47	53	61	71

2 / WHY THE VALUES ARE PRIME / THE LINE MISSES THE HYPERBOLA MODULO p

A prime p divides $f(n)$ exactly when the lattice point $(n+1,-n)$ lies on $uv\equiv k \pmod p$



For $d=163$ and $p=5$ we have $k=41\equiv1$, so the hyperbola $uv\equiv1$ has the squares $(1,1),(2,3),(3,2),(4,4)$ and the line $u+v\equiv1$ has $(0,1),(1,0),(2,4),(3,3),(4,2)$, and the two sets are disjoint, so $5\nmid f(n)$ for every n. The number of crossings in $\mathbb{F}_p^2$ is $1+\left(\frac{-d}{p}\right)$, and the Heegner property is that $-d$ is a nonresidue for every odd prime $p<k$, so there are no crossings below k. Then $f(n)$ has no prime factor below k, and since $f(n)<k^2$ for $0\le n\le k-2$ each value is prime.

$\mathbb{F}_5^2$ for $d=163$

3	5	7	11	13	17	19	23	29	31	37	41	43	47	53
0	0	0	0	0	0	0	0	0	0	0	2	2	2	2

crossings of the line and hyperbola in $\mathbb{F}_p^2$, for each prime p

At $p=41=k$ the hyperbola degenerates to the axes $uv\equiv0$, which the line meets at $(1,0)$ and $(0,1)$, that is $n=40$ and $n=0$, giving $f(40)=41^2$ and $f(0)=41$.

3 / ALL NINE HEEGNER NUMBERS / THE SAME PICTURE WITH $k=\frac{d+1}{4}$

Seven odd Heegner numbers give Euler-type polynomials, and 1 and 2 use the line $u+v=0$

d	(s,P)	Δ	polynomial	consecutive primes	next value
1	$(0,1)$	-4	n^2+1	ring $\mathbb{Z}[i]$	
2	$(0,2)$	-8	n^2+2	ring $\mathbb{Z}[\sqrt{-2}]$	
3	$(1,1)$	-3	n^2+n+1	$f(1)=3,\ f(2)=7,\ f(3)=13$	$f(4)=21$
7	$(1,2)$	-7	n^2+n+2	$n=0$: 2	$f(1)=4=2^2$
11	$(1,3)$	-11	n^2+n+3	$n=0,1$: 3, 5	$f(2)=9=3^2$
19	$(1,5)$	-19	n^2+n+5	$n=0..3$: 5, 7, 11, 17	$f(4)=25=5^2$
43	$(1,11)$	-43	n^2+n+11	$n=0..9$: $11,\dots,101$	$f(10)=121=11^2$
67	$(1,17)$	-67	n^2+n+17	$n=0..15$: $17,\dots,257$	$f(16)=289=17^2$
163	$(1,41)$	-163	n^2+n+41	$n=0..39$: $41,\dots,1601$	$f(40)=1681=41^2$

4 / WHY THE LIST STOPS / COUNTER-EXAMPLES AND THE THEOREM

A line that meets the hyperbola modulo a small prime breaks the run

Take $k=101$, where $\Delta=-403$ is not a Heegner discriminant: $f(4)=121=11^2$, because the lattice point $(5,-4)$ has $5\cdot(-4)=-20\equiv2\equiv101 \pmod{11}$ and so lies on the hyperbola modulo 11. Likewise $k=47$ fails at $f(1)=49=7^2$. The theorem of Baker, Heegner and Stark says exactly nine d have class number one, and Rabinowitsch's criterion says n^2+n+k is prime for $0\le n\le k-2$ exactly when $\mathbb{Q}(\sqrt{1-4k})$ has class number one, which leaves $k=2,3,5,11,17,41$. In the (s,P) plane each Heegner point sits above the parabola $P=s^2/4$ by $d/4$.

ASIDE / THE EVEN CASES

For $d=1$ and 2 the discriminant is $-4d$ and the line is $u+v=0$, through the origin, with hyperbolas $uv=1$ and $uv=2$.

ASIDE / CHECK IN PYTHON

`all(isprime(n*n+n+41) for n in range(40))` gives `True`, and `1681 == 41**2`.

Convention. d is a Heegner number, $k=(d+1)/4$ for odd d, and $\Delta=1-4k=-d$. After the earlier sheets on the Legendre symbol and on lines meeting hyperbolas.